SOCIAL MEDIA ENGAGEMENT ANALYTICS

Python for Data Analysis | One-page project summary

Project overview:

Analyzed 5,000 social media records with 19 fields using Python, NumPy, Pandas, Matplotlib, Seaborn, and Plotly. The work covered data inspection, cleaning, feature creation, statistical summaries, grouping, correlation analysis, and visual exploration.

Data preparation and analysis completed:

• Identified missing values (150 each in age, gender, likes, comments, shares, and sentiment; 3% per affected field) and filled numeric fields
with medians and categorical fields with modes.

• Checked duplicates (none found), standardized text/category labels, converted posting dates to datetime, checked negative values, and
applied IQR based outlier capping.

• Created hashtag count, engagement score (likes + comments + shares), posting date/hour, and engagement-level groups. Used groupby
summaries, concat, functions, loops, sorting, and NumPy array operations.

• Calculated mean, median, mode, standard deviation, variance, and quartiles for key metrics. Produced Matplotlib/Seaborn charts and
three Plotly interactive charts.

Key Findings and Interpretations:

1. Post Type: Video posts recorded the highest average engagement score (12,664.6), followed by image posts.

2. Content Category: Music achieved the highest average engagement score (12,788.5), while Fashion recorded the lowest (12,356.4).

3. Country-wise Engagement: Australia recorded the highest average engagement rate (0.401), whereas India recorded the lowest (0.343).

4. Age Group: Users aged 30-40 recorded the highest average engagement score (12,938.5), indicating greater interaction within this age group.

5. Verified Accounts: Non-verified accounts recorded a slightly higher average engagement score (12,644.3) than verified accounts (12,309.3), suggesting that verification status alone does not guarantee higher engagement.

6. Device Type: Mobile users recorded the highest average watch time (4,087.8 seconds), followed by tablet and desktop users.

7. Sentiment Analysis: Negative sentiment posts achieved the highest average engagement score (12,731.5), while neutral sentiment posts recorded the lowest (12,448.2).

8. Correlation Analysis: Likes and engagement score showed a strong positive correlation ($r = 0.984$), while impressions and engagement rate showed a strong negative correlation ($r = -0.781$).

Conclusion:

The analysis highlights variations in social media engagement based on post type, content category, country, age group, account verification status, device type and sentiment. Video posts and music-related content performed better in terms of average engagement score. Australia recorded the highest average engagement rate, while mobile users showed the highest watch time. The findings demonstrate how Python-based data analysis and visualization can help understand audience behaviour, identify engagement patterns and support data-driven social media content strategies.